him since he was a teenager: No matter how hard you chase after a light beam, it still retreats from you at light speed. You can't make the apparent speed with which light departs one iota less than 670 million miles per hour, let alone slow it down to the point of appearing stationary. Case closed. But this triumph over paradox was no small victory. Einstein realized that the constancy of light's speed spelled the downfall of Newtonian physics.

Truth and Consequences

Speed is a measure of how far an object can travel in a given duration of time. If we are in a car going 65 miles per hour, this means of course that we will travel 65 miles if we persist in this state of motion for an hour. Phrased in this manner, speed is a rather mundane concept, and you may wonder about the fuss we have made regarding the speed of baseballs, snowballs, and photons. However, let's note that distance is a notion about space—in particular it is a measure of how much space there is between two points. Also note that duration is a notion about time—how much time elapses between two events. Speed, therefore, is intimately connected with our notions of space and time. When we phrase it this way, we see that any experimental fact that defies our common conception about speed, such as the constancy of the speed of light, has the potential to defy our common conceptions of space and time themselves. It is for this reason that the strange fact about the speed of light deserves detailed scrutiny—scrutiny given to it by Einstein, leading him to remarkable conclusions.

The Effect on Time: Part I

With minimal effort, we can make use of the constancy of the speed of light to show that the familiar everyday conception of time is plain wrong. Imagine that the leaders of two warring nations, sitting at opposite ends of a long negotiating table, have just concluded an agreement for a cease-fire, but neither wants to sign the accord before the other. The secretary-general of the United Nations comes up with a brilliant resolution. A light bulb, initially turned off, will be placed midway between the two presidents. When it is turned on, the light it emits will reach each of the presidents simultaneously, since they are equidistant from the bulb. Each president agrees to sign a copy of the accord when he or she sees the light. The plan is carried out and the agreement is signed to the satisfaction of both sides.